PROV LOGIC

ULTIMATE ANALOG LAYOUT INTERVIEW GUIDE

Crack Your Dream Job

Repeatedly Asked 50 Interview Questions from Top Product Companies



Prasanthi Chanda

1. What is Latch-Up in CMOS and How Do You Prevent It?

Company: Intel, STMicroelectronics

- **Latch-up: Unwanted parasitic thyristor conduction path between VDD ↔ GND**
- **Triggers: Over-voltage, noise injection, poor well tie**
- **Effects: Sudden high current → chip failure**
- **Prevention Techniques:**
- **Use guard rings (N-well for PMOS, P-sub for NMOS)**
    - **Implement deep N-well isolation**
    - **Maintain sufficient spacing between NMOS/PMOS**
    - **Ensure strong substrate and well contacts**
    - **Use latch-up rules from foundry DRC decks**

2. What is Antenna Effect and How to Fix It in Layout?

Company: Synopsys, GlobalFoundries

- **Happens when metal area > gate area during fabrication steps**
- **Gate terminal collects charge → gate oxide rupture risk**
- **Critical in advanced nodes where metal-to-gate ratios are tight**
- **Fix Methods:**
    - **Add antenna diode (sacrificial path to discharge)**
    - **Use metal jumpers to higher layers early**
    - **Break metal lines with vias or use routing tricks**
    - **Refer to antenna ratio checks in LVS/ERC decks**
    - **Use dummy gates where needed to balance metal-to-gate**

3.What is the Importance of Matching in Analog Layout Design?

Company: Texas Instruments, AMD

- **Matching reduces offset, gain error, and drift in analog blocks**
- **Crucial in: Current mirrors, differential pairs, bandgaps, etc.**
- **Techniques:**
  - **Use common centroid placement**
  - **Ensure identical orientation and proximity**
  - **Match lengths of interconnects**
  - **Use dummy devices at edges**
  - **Avoid gradient-prone corners or hot spots**
- **Tip: Use unit device replication rather than large aspect ratio FETs**

4. How Do You Lay Out a High-Speed Analog Signal Path?

Company: Apple, Qualcomm

- **Priorities: Signal integrity, low parasitics, and matching**
- **Guidelines:**
  - **Use minimum parasitic capacitance (short, wide metals)**
  - **Avoid 90° metal bends – use 45° or curved paths**
  - **Route with symmetry for differential paths**
  - **Keep away from digital/noisy signals (EMI/IR concerns)**
  - **Insert shielding (ground lines) around sensitive nets**
  - **Avoid crossovers or transitions unless necessary**
  - **Use higher metal layers for critical analog signals**

5. How Do You Ensure Good Matching in a Current Mirror Layout?

Company: NXP, Analog Devices

Poor matching leads to current mismatch → impacts biasing and performance.

How to do it:

- **Use common centroid layout**
- **Maintain same orientation (no rotation/mirroring)**
- **Match interconnect lengths and layers**
- **Insert dummy devices on edges**
- **Avoid layout near heat sources or process gradients**
- **Prefer unit transistor repetition instead of wide transistors**

6. Guard Rings in Analog Layout – When & Why to Use?

Company: Texas Instruments, Infineon

Use Case	Type of Guard Ring	Purpose
Around NMOS in P-sub	P+ connected to GND	Substrate noise isolation
Around PMOS in N-well	N+ connected to VDD	Prevent latch-up
Around sensitive circuits	Double ring (P+ & N+)	High immunity from digital noise

7. Why Is Matching Important in Analog Circuits?

Company: Texas Instruments, Skyworks

Matching reduces differential offsets and improves linearity.

Critical Blocks Requiring Matching:

- **Current mirrors**
- **Differential pairs**
- **Bandgap references**
- **Comparators**

Key Concepts to Remember:

- **Process variations cause mismatch in threshold voltage (Vt), mobility (μ)**
- **Better matching ⇒ Better performance ⇒ Higher yield**
- **Layout techniques can minimize physical mismatch even when process isn't perfect**

8. How Should You Layout a Differential Pair?

Company: AMD, Qualcomm

For Perfect Differential Pair Layout:

- **Use common centroid or interdigitated layout**
- **Keep equal routing lengths for both gates**
- **Devices must be identical in orientation (no mirroring!)**
- **Add dummy transistors on outer ends**

Tip: Route differential pair signal traces side-by-side with constant spacing and minimal bends to preserve phase alignment.

9. What Are the Common Parasitics in Analog Layout and Their Impact?

Company: Broadcom, STMicroelectronics

Parasitic	Source	Impact
Parasitic Capacitance (C)	Between metal layers/traces	Slower response, stability issues
Parasitic Resistance (R)	In metal traces & polysilicon	Voltage drop, IR drop
Coupling Capacitance	Between signal lines (cross-talk)	Distortion, noise injection
Substrate Noise	Shared substrate with digital	Degrades analog precision

10. Post-Layout Simulation Differs from Schematic – What Do You Check?

Company: Cadence, Intel

Troubleshooting:

- **Check for parasitic capacitance slowing signal edges**
- **Are devices mismatched or asymmetrically laid out?**
- **Look for missing or incorrect dummy devices**
- **Analyze routing lengths & metal layers**
- **Did layout introduce coupling from digital lines?**
- **Compare netlists (schematic vs extracted)**

Tip: Run LVS + PEX early and often. Don’t wait till tapeout.

11. What Is the Effect of Vt Mismatch and How Can It Be Minimized?

Company: Analog Devices, Texas Instruments

Effect of Vt (threshold voltage) mismatch:

- **Creates offset in differential circuits**
- **Causes gain and bias errors**
- **Degrades linearity and symmetry in analog paths**

How to Minimize:

- **Use identical device geometry**
- **Match orientation, spacing, and proximity**
- **Use common centroid & interdigitated layout**
- **Add dummies at edges to maintain uniform etching**
- **Avoid corners or heat-affected areas in floorplan**

Rule: Vt mismatch can't be "designed away" after layout—prevention is layout-first.

12.How to Route Critical Analog Signals in a Mixed-Signal Environment?

Company: Qualcomm, SkyWater

Do's:

- **Route on higher metal layers to reduce parasitic capacitance**
- **Use symmetrical, shielded routing for differential pairs**
- **Maintain constant spacing between sensitive signals**

Don'ts:

- **Avoid crossing analog with digital lines**
- **Don't use long meandering traces**
- **Never mix clock lines with sensitive analog paths**

13. What Is Common Centroid Layout and Why Is It Preferred?

Company: STMicro, Apple

A placement strategy where transistors are arranged symmetrically around a center point to cancel out gradient effects.

Why it's preferred:

- **Cancels systematic process gradients (like doping or temperature)**
- **Improves matching for analog-critical pairs**
- **Essential for current mirrors, differential pairs, and DACs**

Common Pattern Example:

ABBA, AABBBBAA — ensures that each side sees mirrored process conditions

14. You're Seeing Noise in a Sensitive Analog Node. What Could Be the Cause?

Company: MediaTek, Infineon

Check these first:

- **Nearby digital switching signals causing coupling**
- **Missing or weak guard rings**
- **Substrate coupling via shared wells or poor isolation**
- **Long unshielded routing traces**

Quick Fixes:

- **Insert metal shields or grounded tracks**
- **Add or strengthen guard rings**
- **Move analog layout away from noisy areas in floorplan**

15.What is IR Drop and How Does It Affect Analog Circuits?

Company: Intel, ON Semiconductor

IR Drop = Current × Resistance in power lines

Impact on Analog:

- **VDD droop → gain error, offset drift**
- **Affects biasing accuracy and analog precision**
- **May cause false switching or non-linearity in sensitive circuits**

How to Reduce:

- **Use wider power rails**
- **Add multiple vias for low resistance paths**
- **Route VDD/GND on higher metal layers**

Run an IR drop analysis on post-layout netlist, especially for analog cores.

16. What Are Dummy Transistors and When Should You Use Them?

Company: Microchip, TSMC

Dummy transistors ensure uniform etching and chemical processing at edges of active devices.

When to Use:

- **Around edge of transistor arrays (current mirrors, diff pairs)**
- **In matching-critical blocks**

Layout Tips:

- **Dummies must have same width and gate orientation**
- **Connect gates to GND or VDD (based on design)**

17. What Is the Difference Between Common Centroid and Interdigitated Layout?

Company: Qualcomm, Renesas

Feature	Common Centroid	Interdigitated
Arrangement	Symmetric about center	Alternating finger pattern
Use case	Cancel gradient effects	Improve matching
Pattern	ABBA / AABB...	ABABAB...
Best for	Diff pairs, bandgaps	Matched MOSFETs, resistors

18. Your Comparator Shows Offset After Layout. What Could Be the Issue?

Company: Maxim Integrated, Analog Devices

Debug Path:

- Mismatch in input pair layout (size/orientation)
- Asymmetrical routing to gates or sources
- Long, unmatched gate interconnects
- Poor common centroid practice
- Missing dummy devices

Fix Approach:

- Reroute for symmetry
- Shorten and equalize signal traces
- Add or correct dummy transistors
- Review floorplan placement → ensure low-noise region

19. What Is Latch-Up in CMOS and How Do You Prevent It?

Company: STMicroelectronics, Samsung

Latch-up is a parasitic short circuit between power and ground due to unintended triggering of a PNPN thyristor structure in CMOS.

Trigger causes:

- **Overshoot on I/O pins**
- **Substrate noise or voltage spikes**
- **Improper guard ring isolation**

Prevention Techniques:

- **Use guard rings around NMOS/PMOS**
- **Ensure proper n-well & p-sub spacing**
- **Add substrate/well contacts close to active areas**
- **Follow ESD design rules**

20. What Is the Purpose of Adding Metal Fill in Layout?

Company: GlobalFoundries, TSMC

Metal fill ensures uniform metal density across the chip for proper planarization and to meet CMP (Chemical-Mechanical Polishing) requirements.

Things to Keep in Mind:

- **Avoid placing metal fill too close to sensitive analog nodes**
- **Use dummy fills to balance density in analog vs digital regions**
- **Can act as parasitic coupling paths — isolate with grounded shielding if needed**
- **Metal fill must meet DRC and foundry rules**

21. What Are Key Differences Between Analog and Digital Layout?

Company: NXP, Broadcom

Analog Layout	Digital Layout
Manual, precision-driven	Mostly automated (place & route)
Focus on matching, noise, symmetry	Focus on density and timing
Uses guard rings, shielding, dummies	Emphasizes clock trees, scan chains
Sensitive to parasitics, gradients	More robust to layout imperfections

22. A Bandgap Reference Has Shifted Output After Tape-Out. What Do You Check?

Company: Dialog Semiconductor, Texas Instruments

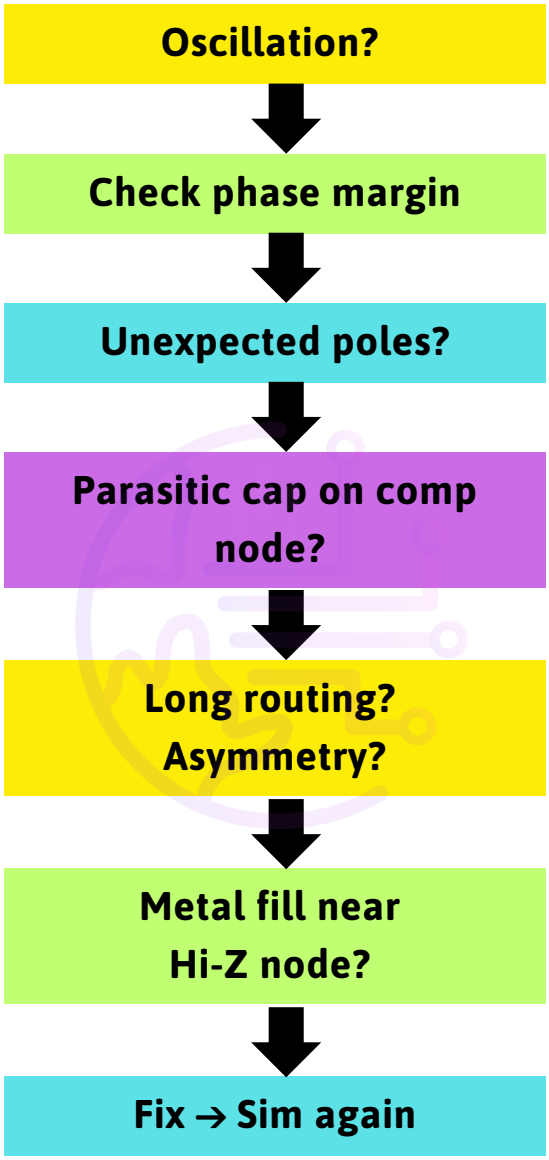
Post-layout debugging checklist:

- **Check mismatch in core current mirrors**
- **Look for parasitic metal capacitance on reference node**
- **Verify symmetry in the layout of bipolar transistors (if used)**
- **Check for IR drop in VDD or GND routing**
- **Inspect any metal fill or proximity effect from neighboring blocks**

Bandgaps are extremely sensitive—90% of failures link back to poor layout handling.

23. Post-Layout, Op-Amp Is Oscillating. How Do You Debug?

Company: Infineon, Marvell



24. What Are the Key Parasitics in Analog Layout and Why Do They Matter?

Company: Infineon, Marvell

Top Parasitics to Watch For:

- **Capacitance (C): At sensitive nodes → affects bandwidth, phase margin**
- **Resistance (R): In routing → causes IR drop, delay, gain error**
- **Coupling (Crosstalk): Between signal and clock or digital lines**
- **Substrate Noise Paths: Via shared wells or poor guard ring layout**

Impact on Circuit:

- **Changes pole/zero locations**
- **Adds offset, noise, or distortion**
- **Reduces matching → hurts performance**

25. Case Study – OTA Output Swing Is Lower Than Expected

Company: Infineon, Marvell

After layout, the OTA (Operational Transconductance Amplifier) can't swing close to VDD or GND. Expected rail-to-rail, but output range is limited.

Root Cause Analysis:

- **Routing added resistive drops on output node**
- **Output connected via long, narrow metal path**
- **No shielding from digital logic — coupling noise seen**
- **Output stage had mismatched load devices due to bad symmetry**

Fixes Applied:

- **Re-routed with wider metal layers (Metal 3+)**
- **Shortened output path → placed closer to load**
- **Used common centroid layout on output load**
- **Added guard ring + shield metal around output**

Takeaway:

Even if schematically correct, poor layout kills swing. Analog layout isn't just wires — it's circuit performance carved in metal.

26. Trap vs. Best Practice – Laying Out Differential Pairs

Company: AMD, Synaptics

Trap	Best Practice
Placing transistors with same W/L but different orientation	Mirror or rotate identically for symmetry
Routing gates with different metal layers/lengths	Use identical path and metal layers
No dummy devices at edges → mismatch	Add dummies to maintain process uniformity
Mismatch in current mirror layout feeding diff pair	Use common centroid + interdigitated layout
Placing pair near digital blocks	Isolate with analog guard rings & spacing

27. OTA Shows Phase Margin Drop Post-Layout. What to Suspect?

Company: Broadcom, OnSemi

Potential Issues:

- **Extra parasitic capacitance on compensation node**
- **Long routing on feedback or output node**
- **Asymmetry in load transistors**
- **Routing resistance → shifting poles**
- **Overlapping metal fill near sensitive node**

Action Plan:

- **Extract parasitics → run AC & transient sim**
- **Reroute with short, wide tracks**
- **Add shielding metal**

28. Do’s and Don’ts – ESD Protection Layout

Company: Texas Instruments, NXP Semiconductors

Do’s	Don’ts
Use dedicated ESD diodes at all pads	Don’t rely on internal clamp only
Place ESD cells as close as possible to pads	Don’t route long wires from pad to clamp
Follow foundry ESD layout rules	Don’t skip min spacing/width checks
Use wide, multi-via GND/VDD connections	Don’t use thin wires for ESD discharge
Add guard rings around clamps	Don’t place ESD near sensitive analog paths

29. How Do You Lay Out a High-Speed Comparator Block?

Company: Qualcomm, Realtek

Step 1 – Understand the Block

- **Inputs are fast-changing → noise-sensitive**
- **Speed depends on matching and low parasitics**
- **Output drives digital logic — watch for crosstalk**

Step 2 – Floorplan Wisely

- **Place preamp, latch, and output buffer in sequence**
- **Keep critical paths short & shielded**

Step 3 – Layout the Input Stage

- **Use common centroid for input differential pair**
- **Match gate routing**
- **Add dummies and use double contacts**

Step 4 – Route for Speed

- **Route output with wide metals**
- **Avoid vias on signal path**
- **Place decaps nearby for supply stability**

Step 5 – Shield & Isolate

- **Guard rings around input stage**
- **Ground shields on input nets**
- **Isolate from clock buffers or digital lines**

30. Why Are Layout Corners (TT, SS, FF, etc.) Simulated Post-Layout?

Company: Analog Devices, MediaTek

Purpose of Corners:

- **Represent process variations (transistor speed & behavior)**
- **Ensure design meets specs across all conditions**

Why After Layout?

- **Layout introduces parasitics that shift performance**
- **Corner + parasitic simulation shows worst-case behavior**

Common Corners:

- **TT (Typical-Typical)**
- **SS, FF (Slow/Slow, Fast/Fast)**

31. Comparator Delay Too High Post-Layout – What Changed?

Company: Microchip, Nuvoton

Before Fix (Problem Layout)

- **Input pair layout was not symmetrical**
- **Gate routing used different metal layers**
- **Comparator placed far from clock buffer**
- **No shielding — noise picked up from nearby clk**

After Fix (Optimized Layout)

- **Used common centroid layout for input pair**
- **Matched routing with same metal and length**
- **Routed output in Metal 3, wider path**
- **Added ground shield and guard ring**

Result:

- **Delay improved by 30%**
- **Reduced noise-induced meta-stability**
- **Better clock alignment → no setup/hold violations**

32. Explain the Role of Guard Rings in Analog Layout

Company: Skyworks, STMicroelectronics

Purpose of Guard Rings:

- **Shields sensitive circuits from substrate noise**
- **Collects leakage current away from devices**
- **Isolates blocks powered by different supplies**

Design Tips:

Use P+ guard ring tied to GND for NMOS blocks

Use N+ guard ring tied to VDD for PMOS blocks

33. How Do You Model Parasitics in Analog Simulations?

Company: NXP, Infineon

Methods:

- **Run Parasitic Extraction (PEX) → Extract R, C (sometimes L)**
- **Use RC-netlists → include in post-layout simulation**
- **Simulate across corners + temperatures**
- **Use Monte Carlo with parasitics for mismatch check**

Why It Matters:

- **Impacts gain, delay, phase margin**
- **Slows down fast nodes**
- **Can kill matching in current mirrors and diff pairs**

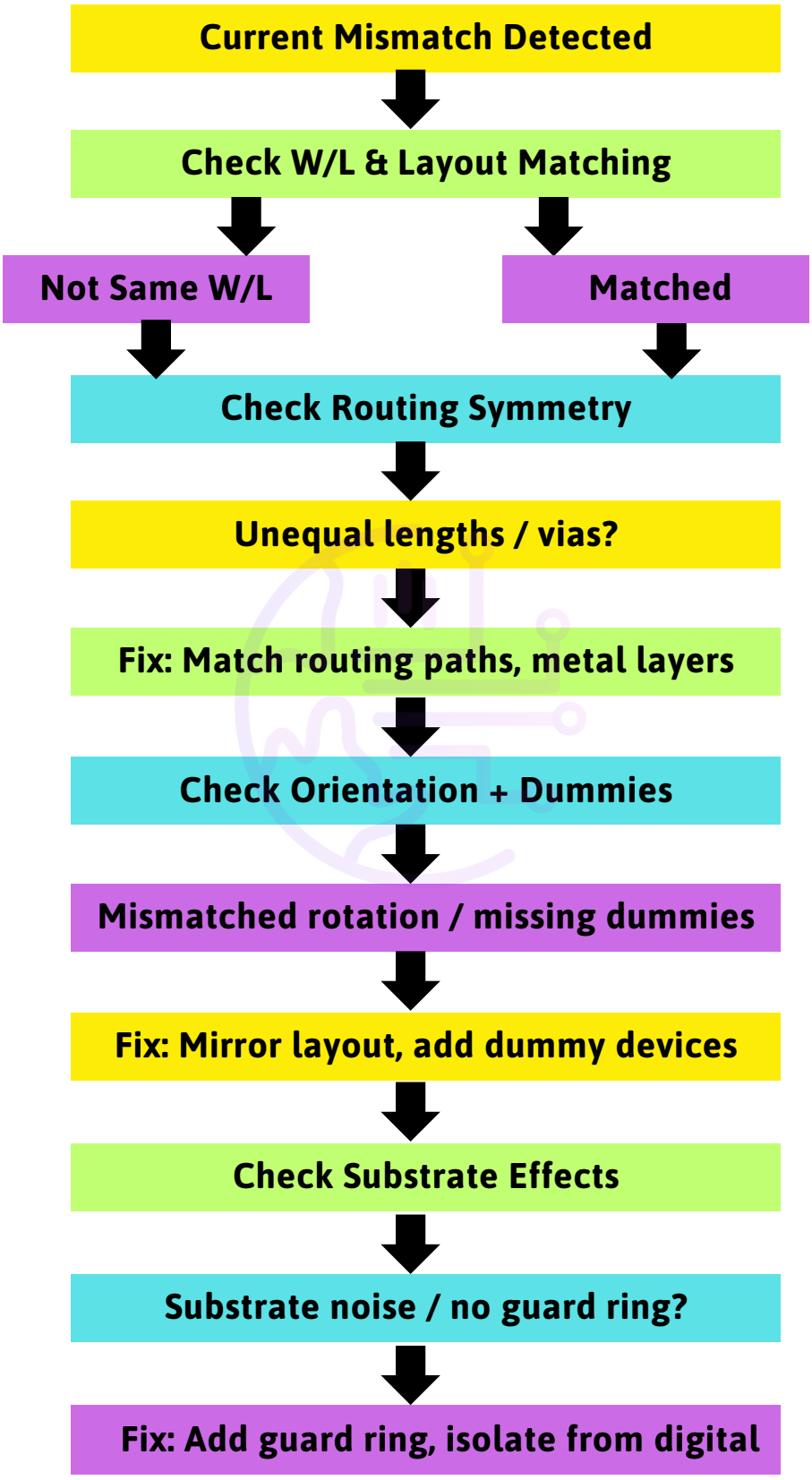
34. Quiz Me Quick: Analog Layout Fundamentals

Company: Texas Instruments, STMicroelectronics

- **What's the difference between metal fill and dummy poly?**
  - **Metal fill = density compliance (post-routing),**
  - **Dummy poly = improves uniform etch rate near active devices.**
- **Why place decaps near supply rails?**
  - **To suppress local supply noise, reduce IR drop, and stabilize fast switching nodes.**
- **When should you avoid using minimum width metal?**
  - **On power rails, analog bias lines, clock nets, or high current paths — increases resistance + IR drop.**
- **What's better for matching: side-by-side or interdigitated layout?**
  - **Interdigitated — reduces gradient & stress mismatch across die.**

35. Flowchart – Debugging a Mismatch in Current Mirror Post-Layout

Company: GlobalFoundries, Samsung Semiconductor



36. What Is Latch-Up? How Do You Prevent It in Layout?

Company: NXP, Infineon

A parasitic thyristor turns ON in CMOS due to noise, ESD, or substrate current → causes short between VDD and GND → chip damage or reset.

Prevention in Layout:

- **Use guard rings to cut off parasitic paths**
- **Ensure adequate well and substrate contacts**
- **Keep VDD/VSS separation where needed**
- **Follow foundry latch-up rules**
- **Avoid routing high-current signals near sensitive analog nodes**

37. How Would You Floorplan an Analog Front-End (AFE) Block?

Company: Texas Instruments, Maxim Integrated

- **Understand the Block Flow**
  - **→ Signal path: Sensor → PGA → ADC**
  - **→ Includes bias gen, references, filters**
- **Place High-Gain & Sensitive Blocks First**
  - **→ Keep PGA & bias gen away from digital**
  - **→ Isolate with guard rings**
- **Centralize Bias Distribution**
  - **→ Route short, shielded bias lines to all sub-blocks**
- **Decide Power Routing Early**
  - **→ Wide metals, multiple vias**
  - **→ Local decaps near each block**
- **Clock/ADC Blocks**
  - **→ Place closer to digital boundary**
  - **→ Shield analog interface lines from cross-talk**

38. What Is Effective Channel Length? How Does It Affect Analog Design?

Company: Analog Devices, Renesas

Effective Length (Leff) = drawn length – lateral diffusion

Actual region under gate where channel forms

Impact in Analog Design:

- **Impacts gm (transconductance)**
- **Impacts ro (output resistance)**
- **Affects matching & gain**
- **Short Leff → faster but higher 1/f noise**
- **Longer Leff → better linearity & output swing**

39. Myth vs Reality: Analog Layout Edition

Company: Qualcomm, Broadcom

Myth	Reality
Matching depends only on W/L	Matching depends on layout symmetry, routing, environment, and dummy placement.
More metal = better routing	Excess metal increases capacitance → slower signal. Use smart, short, wide routing.
Guard rings are optional	Guard rings are essential for isolation, latch-up prevention, and substrate noise control.
Post-layout simulation = formality	Post-layout sim with parasitics is critical for final performance tuning.

40. A Differential Amplifier Shows Offset – What Could Be the Reasons?

Company: STMicroelectronics, MediaTek

Possible Root Causes:

- **Input pair not placed in common centroid**
- **Mismatch in routing parasitics (metal imbalance)**
- **Stress gradient due to poly/STI underneath**
- **Missing or asymmetric dummy transistors**
- **Uneven guard ring layout → substrate coupling**

Fix Actions:

- **Re-layout input pair with common centroid**
- **Match routes with same metal layers + vias**
- **Add dummy poly + diffusion**
- **Re-check floorplan symmetry**
- **Simulate with extracted parasitics**

41.Real Debug Case: Reference Voltage Drifts on Final Silicon

Company: Analog Devices, Infineon

Problem Statement:

- **Bandgap reference output: Expected 1.2V, but measured 1.27V**
- **Temp variation causes additional 20mV drift**
- **Performance degrades in low-temp corners**

Debug Investigation:

- **Checked schematic — all clean, matches golden sim**
- **Post-layout sim matches schematic (PEX netlist OK)**
- **No ESD clamp leakage, power rail stable**

42. How Does Temperature Affect Layout Matching?

Company: Maxim Integrated, Skyworks

Impact of Temperature:

- Increases mobility mismatch
- Affects Vt shift, especially in poly stress layouts
- Causes uneven expansion → layout stress gradient
- Impacts resistor tempco → voltage drop drift
- Can magnify parasitic effects (cap, R)

How to Minimize:

- Use long-channel devices for better thermal stability
- Apply symmetry + dummy devices
- Avoid layouts over poly edges, STI, or corners
- Simulate over TT/SS/FF corners with temp sweep
- Use thermally matched resistor types

43.Routing Challenge: Clean Signal Path in Mixed-Signal SoC

Company: Broadcom, Qualcomm

Challenges:

- Crosstalk from digital logic
- Noise from clock and power supply
- Signal path is long (~200µm)

What Should You Do?

- Route signal on top metal (Metal 4/5) for minimum parasitics
- Add ground shielding (GND-GND-SIGNAL-GND stack)
- Use wide spacing from digital signals

44. Why Is Symmetry So Important in Analog Layout?

Company: NXP, TI, Synopsys AMS

Prevents systematic mismatch

- **Equalizes stress gradients (poly/STI, metal loading)**
- **Ensures common-mode rejection in diff pairs**
- **Helps in matching gain, biasing, noise**
- **Keeps performance across corners + temp**

Where You Apply It:

- **Differential pairs**
- **Current mirrors**
- **Cascode stacks**
- **Bandgap blocks**
- **Matched resistors or caps**

45.Common Layout Mistakes That Lead to Tapeout Failures

Company: Intel Custom Foundry, SkyWater Tech

Mistake	Result / Issue
Forgot dummy poly on analog input pair	Offset, mismatch, temp drift
Routed analog signal near switching clock net	Signal distortion / jitter
Incomplete guard ring or tap tie	Substrate noise coupling
Antenna rule violations missed	Gate oxide damage at first power-on
Power rail routed with minimum width	IR drop, analog block instability

46. What Is Antenna Effect and How Do You Fix It?

Company: GlobalFoundries, Renesas

Antenna Effect = Plasma-Induced Damage

- **Happens when long metal traces are left unconnected to diffusion during fabrication**
- **Charge builds up → gate oxide breakdown**
- **Especially severe in analog gates with long M1/M2 segments**

Fixing It in Layout:

- **Use antenna diodes (connect metal to diffusion early)**
- **Break long metal segments and connect to vias**
- **Route signals closer to diffusion contact points**
- **Use jump connections to anchor metal early**
- **Follow foundry antenna rules strictly (e.g., area ratio)**

47. Design Review Simulation: Defend Your Layout

Company: Apple Silicon, AMD

You just completed layout for a Low-Noise Amplifier (LNA) block. You're in a design review with senior analog leads. You are asked the following:

Reviewer 1:
Why did you place the input diff pair on the top-left corner of the block?
Expected Answer:

- **Minimized parasitics by keeping input near pad**
- **Allowed clean routing of signal shield**
- **Avoided noisy blocks (bias gen, power switches)**

Reviewer 2:

Did you use any symmetry strategy?

Expected Answer:

- **Yes — Common centroid for the input pair**
- **Routing mirrored + matched (metal 2 top, shielded bottom)**
- **Dummies placed for matching edge effects**

Reviewer 3:

How did you route the bias lines?

Expected Answer:

- **Routed bias on top metal (M4) — short and shielded**
- **Used wide metal with matched path lengths**
- **Avoided running bias over digital or poly edges**
- **Used GND shielding to suppress coupled noise**

48. What Is Common Centroid Layout? Why Is It Used?

Company: Texas Instruments, Dialog Semiconductor

Common Centroid Layout:

- **A placement method where matched devices are split and interleaved**
- **Geometry arranged so any gradient (temp, stress, dopant) affects both sides equally**

Why It's Used:

- **Improves matching accuracy**
- **Cancels linear gradients**
- **Used in diff pairs, current mirrors, capacitors**
- **Minimizes offset and gain errors**

49. What Layout Strategies Do You Use to Minimize 1/f Noise in Analog Blocks?

Company: Analog Devices, NXP

What Causes 1/f Noise:

- **Dominant in low-frequency analog designs**
- **Originates from traps at oxide interface**
- **Strong in MOSFETs (especially NMOS)**

Layout Strategies to Reduce It:

- **Use PMOS over NMOS (PMOS has lower 1/f noise)**
- **Increase device area (W×L) → more averaging**
- **Avoid STI edges / poly bends under active region**
- **Place input devices on symmetrical layout**
- **Use guard rings + shielding for substrate noise control**
- **Add dummy gates and diffusion to balance stress**

50. How Do You Layout a High-Speed Comparator in a Noisy Environment?

Company: Broadcom, STMicroelectronics

Challenges:

- **Comparator is sensitive to kickback noise, supply bounce**
- **Needs low offset, fast transition, clean edges**
- **Operates near sub-mV input swing**

Layout Must-Haves:

- **Input pair on common centroid + shielded**
- **Guard rings around input & tail current source**
- **Metal shielding + routing symmetry**
- **Keep noisy digital block physically away**
- **Use deep Nwell / triple well to isolate substrate**

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